

Laboratory assignment

Component (1)

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1 Overview

For the first software component of this software project, the chosen MAS (Multi-agent System) framework is SPADE ("**Smart Python Agent Development Environment**").

2 SPADE Framework

2.1 Programming Language

SPADE is written entirely in Python. It relies on Python version 3.8+ and "**Asyncio**" Python library (a library to write concurrent code using the `async/await` syntax).

3 Framework Features

Its key features include:

- **Agent Design:** The framework enables the definition and usage of agent classes. Every agent is assigned a unique identifier (e.g. `my_agent@server.com`);
- **Behaviour-Based Model:** Define agent behaviors using cyclic, periodic, one-shot and finite state machine patterns. A behavior is a specific task or job that the agent performs;
- **Concurrent/Parallel Execution:** Thanks to the "**Asyncio**" library, the framework enables asynchronous programming and concurrent agent operations;
- **Communication:** The framework is built on the XMPP ("**Extensible Messaging and Presence Protocol**") protocol for robust, real-time messaging between agents and humans. This is a standard internet technology originally designed for human chat applications. The effective usage of this feature relies on a running XMPP server, such as "**Openfire Server**";
- **Presence Awareness:** Allows for the monitoring of agent status and availability through a real-time presence notification system;
- **Web Interface:** Provides a built-in web interface for easy monitoring and interaction with your agents;
- **LLM Integration:** It's newest feature allows seamless integration with Large Language Models for intelligent agent reasoning and natural language processing.